

TOPIC SIX: MECHANICAL TRANSPORTATION OF PEOPLE AND GOODS IN BUILDINGS.

General Introduction.

In modern construction, many developers are going vertical to utilize space. Buildings today are taller, busier, and more crowded than ever. People want fast movement, comfort and safety.

High-rise buildings will have a huge challenge in movement of people and goods up and down their floors. In fact, it will be against general health and safety of people. It is therefore necessary for buildings to have mechanical transportation of people and goods in buildings. Without proper vertical movement, your building becomes slow, stressful, and unattractive. When you install both lifts and escalators, you:

- Improve traffic flow
- Reduce congestion
- Increase property value
- Improve user experience
- Boost rental and sales demand

Any building with more than four floors must have a lift as stipulated in the Building Code.

For movement of people and goods, a building can have one or a combination of the following mechanical transportation installations:

- Lifts
- Travelators
- Escalators
- Hoists

Regulatory Framework in Kenya

Mechanical transport systems are governed by:

- **Kenya Building Code (2024 draft / legacy code)**
- **National Construction Authority (NCA)** guidelines
- **Kenya Bureau of Standards (KEBS)** – safety & equipment standards
- Fire safety provisions by County Governments
- The **Mechanical and Transport** Division (MTD)-responsible for provision, maintenance and renewal of **mechanical transport** and Plant services

Key requirement:

- Buildings **above 4 floors MUST have lifts**

- Buildings above **18m height require firefighting lifts**

Primary functional requirements for mechanical transportation installations

- Safety:** Safety is paramount. Lifts, escalators and moving walkways must be designed and tested to ensure the highest safety standards. If well maintained and serviced, these mechanical transport systems should provide relatively trouble-free and safe transportation. Lifts must not be used in a fire; however, 'firefighting' lifts can be used by firefighters.
- Reliability and maintenance:** Given that lifts and escalators are crucial to the ease of circulation of people between different floor levels, the reliability and the quality of the after-sales service are an important consideration for building designers and building owners. Having mechanical transport systems out of order for a very short period of time is inconvenient, and a lengthy shutdown can be expensive in terms of loss of business and disruption to staff and customers.
- Many commercial buildings will require 'immediate' responses from the manufacturer's service department. Before a final choice of manufacturer is made, the proximity of the manufacturer's service branches to the building should be checked
- Quiet and smooth operation:** The finishes of the lift car, the lift speed and the smoothness of the ride will affect the experience of using the lift, escalator or walkway. Modern lifts are both quiet in operation and provide a smooth delivery to the required floor. Traction lifts tend to provide a gentler and faster ride than hydraulic lifts. Escalators tend to generate a small amount of noise during operation.
- Speed:** The comfort level of those using the apparatus limits the speed of escalators and walkways. The speed is restricted to provide a safe environment to step onto and off the escalators and moving walkways. Traction lifts are often marketed on their speed, especially in high-rise buildings. Hydraulic lifts have a maximum moving speed of 1 m/s. Fastest traction lifts are capable of speeds of around 6 m/s, although more typically the speed is around 1.6 m/s.
- Aesthetics:** The interior finish of the lift car and lighting, the ease of use of the call buttons, and finish of the landing doors all contribute to the overall experience of using the lift. With escalators and walkways, the handrail/balustrade and surface finish contribute to the aesthetic of the apparatus.
- Coordination and tolerances:** An essential requirement when designing and planning buildings that utilize lifts and escalators is the coordination of dimensions.

Lifts and escalators are manufactured to precise dimensions, and manufacturers set out specific requirements for the amount of tolerance in the built structure that supports and/or encloses their equipment. Failure to coordinate dimensional drawings prior to construction, and especially failure to liaise regarding any changes that occur during construction, may result in considerable rework on site. Critical dimensions are the finished floor to finished floor dimensions, the internal size of lift shafts, the width and height of the structural opening to the lift shaft, the pit depth (ground floor level to bottom of pit) and the headroom (height from the top of the lift car exterior to the top of the lift shaft). The lift shaft must be constructed plumb and in accordance with the vertical tolerances provided by the lift manufacturer. The formwork and finished concrete shaft must be regularly checked as work proceeds to ensure the work is within the specified tolerance.

Factors governing the selection of mechanical equipment in a building:

- Type of occupancy
- Size and distribution of building population
- Number of storeys/floors
- Floor shape and dimensions
- Building regulations

1. LIFTS

A lift (an elevator) is a hoisting or lowering mechanism, designed to carry passengers or freight, and is equipped with a car and platform that typically moves in fixed guides and serves two or more landings.

A. Classification/types of lifts.

Classifications of lifts can be done based on:

I. Means of raising or lowering them:

The elevators can be broadly classified as either electric traction type or hydraulic type.

- a) **Traction elevators** have an elevator car and counterweight attached to opposite ends of hoist ropes. The hoist ropes pass over a driving machine that raises and lowers the car. Traction elevators run on load-bearing rails in the elevator hoist way. Traction elevators are most often used in mid-rise and high-rise buildings with five or more floors.
- b) **Hydraulic elevators**, on the other hand, are raised by forcing pressurized oil through a valve into a steel cylinder located above ground or underground. The pressure forces

a piston to rise, lifting the elevator platform and car enclosure mounted on it. The car is lowered by opening the valve and allowing the weight of the car to force oil from the cylinder in a controlled manner. When the valve is closed the car is stopped. Since the weight of hydraulic elevator cars is borne by the piston, there is no need for a structural framework or load-bearing rails. Hydraulic elevators are commonly found in low-rise buildings with two to five floors. Hydraulic lifts offer a number of benefits over traction lifts. Life cycle maintenance costs may be lower, simply because there are less pulleys and lengths of wire ropes, hence less wear. A hydraulic lift uses power only in the ascent; it uses gravity to descend and a small amount of energy to operate the valves. Energy savings may be possible, compared with traction lifts. The hydraulic oil has a long lifespan and will usually last the life of the lift. If the oil needs to be replaced, it can be recycled.

II. Classification based on use

- a) **Platform lifts** - Platform lifts can take a number of different forms, but they are usually used in low-rise buildings where they will only travel a few floors at most. This means they tend to move more slowly than passenger lifts and are often used for disabled access in buildings where most people will take the stairs.
- b) **Passenger lifts** -Passenger lifts are designed to carry people and can come in a variety of forms. As these lifts carry passengers, they should meet specific requirements and standards to ensure that they are safe. They can come in a range of sizes, transporting different numbers of passengers.
- c) **Service lifts**-Service lifts tend to not be designed to carry passengers, but usually carry goods of some description. This also means that these lifts may not have to meet the same level of standards required of passenger lifts.
- d) **Dumb waiters** -Dumb waiters are often used in a kitchen setting, in restaurants, bars, hotels, etc. These small lifts are used to transport food from the kitchen to a serving area. They can also be used in other settings such as hospitals or offices.
- e) **Trolley lifts** -Trolley lifts are designed to transport larger goods on trolleys and roll cages. They are often found within a shop setting or any other environment where you may need to take deliveries and transport them between floors.
- f) **Disabled access lifts** - Disabled access lifts can take a variety of forms and can fall under many different categories, while serving the purpose of transporting those with mobility issues.

- g) **Stair lifts** -Stair lifts are most commonly used in the home and provide an easy way for individuals to move up and down stairs. This usually involves them sitting in a motorised seat which takes them from one level to another. This, however, is not suitable for individuals who are confined to a wheelchair.
- h) **Evacuation lift** -In general it is not appropriate to use lifts when there is a fire in the building because there is always the danger of people being trapped in a lift that has become immobilised as a result of the fire. However, in some circumstances a lift may be provided as part of a management plan for evacuating people.
- i) **Firefighting lift** - A lift designed to have additional protection, with controls that enable it to be used under the direct control of the fire and rescue service in fighting a fire. A firefighting lift is required if the building has a floor more than 18m above, or more than 10m below fire service vehicle access level.

B. GENERAL LIFT PLANNING

Planning (as with all services) should commence early in the design programme. Priority must be given to locating lifts centrally within a building to minimize horizontal travel distance. Consideration must also be given to position, relative to entrances and stairs. Where the building size justifies several passenger lifts, they should be grouped together. In large buildings it is usual to provide a group of lifts near the main entrance and single lifts at the ends of the building. The lift lobby must be wide enough to allow pedestrian traffic to circulate and pass through the lift area without causing congestion. For tall buildings in excess of 15 storeys, high speed express lifts may be used which by-pass the lower floors.

Planning Considerations

Several factors combine to influence the planning and designing of lift installation, including the passenger handling capacity, waiting interval, speed, location, finishes, intelligent group control safety, and reliability. There are also risks associated with the use of elevators. To ensure that persons are not stuck in elevators for longer periods of time, or worse that the elevator does not lose stability and plummet to the basement from a high floor, the engineers responsible for designing elevators must comply with all statutory codes and standards.

Typical parameters in design of lifts include:

- i. **Characteristic of the premises**
 - Type and use of building;
 - Floor plate size and height of the building;
 - Size of population and its distribution in the premises;
 - Fire safety and regulations;

- The house keeping of the premises.

ii. Circulation Efficiency

- Number of cars and their capacity;
- Location and configuration of elevators in entrance lobby;
- Travel length, number of stops and maximum acceptable waiting time;
- Arrangement with the combination of elevator, escalator and emergency stairs.

iii. Characteristic of the lift equipment

- Type of transportation systems;
- Rated load and car dimensions;
- The speed of the lift/escalator system;
- The type of motor drive control system of the machine;
- Mode of group supervisory control and safety features;
- Cab enclosure and hoist way door finishes;
- Emergency power supplies and fire protection systems;
- Requirements of the local regulations on vertical transport system.

C. Control of lifts

- **Single Automatic Lift Control:** The lift car can be called and used by only one person or group of people at a time. When the lift car is called to a floor, the signal lights engraved 'in use' are illuminated on every floor. The car will not respond to any subsequent landing calls, nor will these calls be recorded and stored. The car is under complete control of the occupants until they reach the required floor and have departed the lift. The 'in use' indicator is now switched off and the car is available to respond to the next landing call. Although the control system is simple and inexpensive by comparison with other systems, it has its limitations for user convenience. It is most suited to light traffic conditions in low rise buildings such as nursing homes, small hospitals and flats.
- **Down Collective Lift Control:** Down collective - stores calls made by passengers in the car and those made from the landings. As the car descends, landing calls are answered in floor sequence to optimize car movement. If the car is moving upwards, the lift responds to calls made inside the car in floor sequence. After satisfying the highest registered call, the car automatically descends to answer all the landing calls in floor sequence. Only one call button is provided at landings. This system is most

suited to flats and small hotels, where the traffic is mainly between the entrance lobby and specific floors.

- **Full or directional collective:** This is a variation in which car and landing calls are immediately stored in any number. Upward and downward intermediate landing calls are registered from one of two directional buttons. The uppermost and lowest floors only require one button. The lift responds to calls in floor order independent of call sequence, first in one direction and then the other. It has greater flexibility than the down collective system and is appropriate for offices and departmental stores where there is more movement between intermediate floors.
- **Controls for Two or More Cars:** Two cars may be coordinated by a central processor to optimize efficiency of the lifts. Each car operates individually on a full or down collective control system. When the cars are at rest, one is stationed at the main entrance lobby and the other, which has call priority, at a mid-point within the building or at another convenient floor level. The priority car will answer landing calls from any floor except the entrance lobby. If the priority car is unable to answer all call demands within a specific time, the other car if available will respond. A similar system may also apply to three cars, with two stationary at the entrance lobby and one available at mid-point or the top floor.

D. Lift Safety Features

- **Buffers** - located at the base of the shaft. They are usually oil loaded for lift speeds >1.5 m/s and otherwise spring loaded. Some variations use compressible plastics.
- **Overspeed governor** - a steel rope passes round a tension pulley in the pit and a governor pulley in the machine room. It also attaches to the lift car's emergency braking system. Over speeding locks the governor as it responds to spring loaded fly-weight inertia from the centrifugal force in its accelerating pulley. This also switches off power to the lift. The tightening governor rope actuates the safety braking gear.
- **Safety gear** - hardened steel wedges are arranged in pairs each side of the lift car to slow down and stop the car by frictional contact with the car guide rail. Slow- and medium-speed lifts have pairs of hardened steel cams which instantaneously contact a steel channel secured to the lift hwall.

Lift performance depends on:

- Acceleration;

- Retardation;
- Car speed;
- Speed of door operation; and
- Stability of speed and performance with variations of car load.

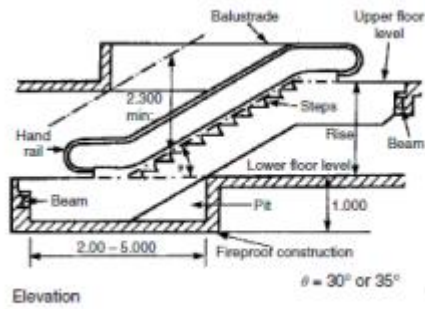
F. Lift features that enhance the comfort and convenience of lift users:

- Directional indication of location of the lift lobby for people unfamiliar with the building.
- Call buttons at landings and in the car positioned for ease of use with unambiguous definition for up and down directions.
- Call buttons to be at a level appropriate for use by people with disabilities and small children.
- Call display/car location display at landings to be favorably positioned for a group of people to watch the position of all cars and for them to move efficiently to the first car arriving.
- Call lights and indicators with an audible facility to show which car is first available and in which direction it is travelling.
- Lobby space of sufficient area to avoid congestion by lift users and general pedestrian traffic in the vicinity.

2. ESCALATORS

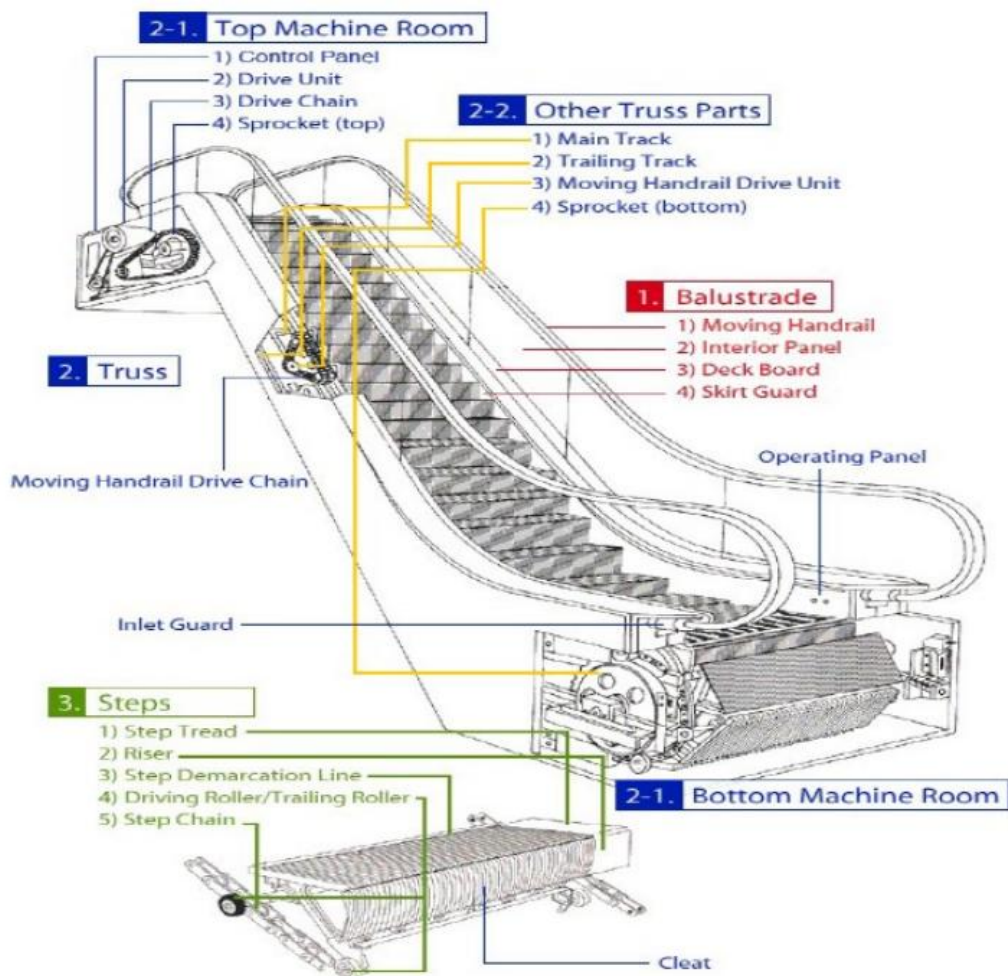
Escalators are moving stairs used to convey people between floor levels. They are usually arranged in pairs for opposing directional travel to transport up to 12 000 persons per hour between them.

The maximum carrying capacity depends on the step width and conveyor speed. Standard steps widths are 600, 800 and 1000 mm, with speeds of 0.5 and 0.65 m/s. Control gear is less complex than that required for lifts as the motor runs continuously with less load variations. In high rise buildings space for an escalator is unjustified for the full height and the high speed of modern lifts provides for a better service.



Components of an escalator

- Landing platforms
- Truss
- Tracks
- Steps
- Handrail



Landing platforms

These two platforms house the curved sections of the tracks, as well as the gears and motors that drive the stairs.

The top platform contains the motor assembly and the main drive gear, while the bottom holds the step return idler sprockets

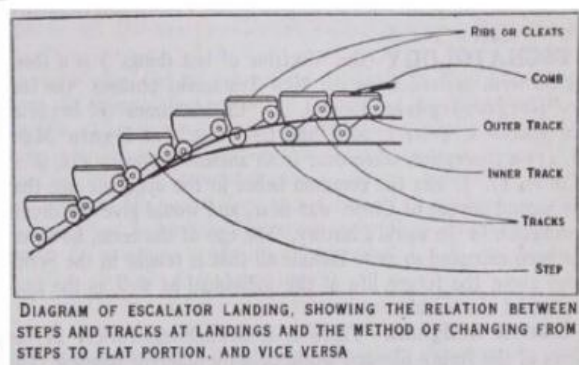


Truss

- The truss is a hollow metal structure that bridges the lower and upper landings.
- It is composed of two side sections joined together with cross braces across the bottom and just below the top. The ends of the truss are attached to the top and bottom landing platforms via steel or concrete supports.

Tracks

- The track system is built into the truss to guide the step chain, which continuously pulls the steps from the bottom platform and back to the top in an endless loop.
- There are actually two tracks: one for the front wheels of the steps (called the step wheel track) and one for the back wheels of the steps (called the trailer- wheel track)



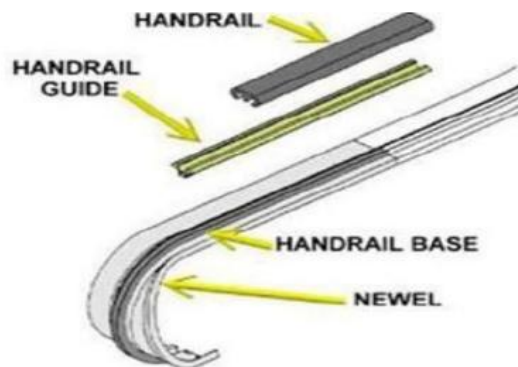
Steps

- The steps are solid, one piece, die-cast aluminum or steel. Yellow demarcation lines may be added to clearly indicate their edges.



Handrail

- The handrail provides a convenient handhold for passengers while they are riding the escalator.



Escalator design

Physical factors

- Physical factors like the vertical and horizontal distance to be spanned must be considered.
- These factors will determine the pitch of the escalator and its actual length.

Location

- Escalators should be situated where they can be easily seen by the general public.
- Furthermore, up and down escalator traffic should be physically separated and should not lead into confined spaces.

Traffic patterns

- Traffic patterns must also be anticipated in escalator design.
- In some buildings the objective is simply to move people from one floor to another, but in others there may be a more specific requirement, such as funneling visitors towards a main exit or exhibit

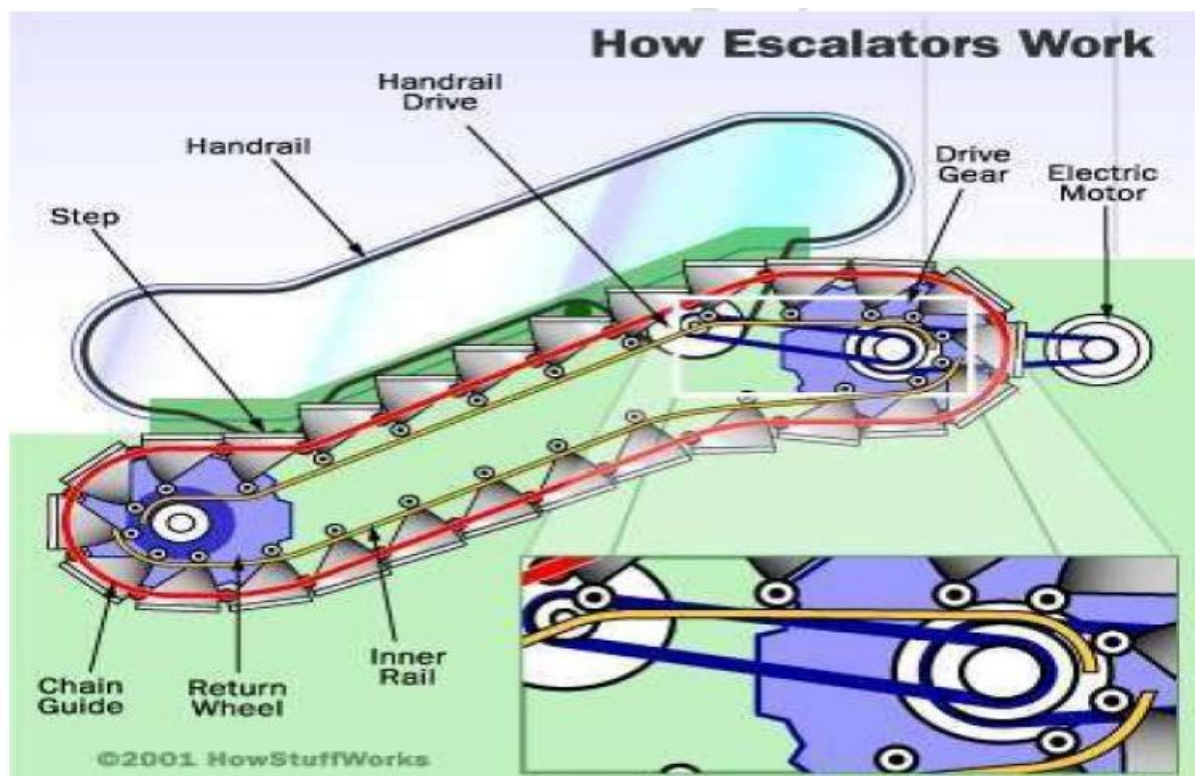
- The number of passengers is important because escalators are designed to carry a certain maximum number of people.

Safety

- Safety is also major concern in escalator design.
- Fire protection of an escalator floor-opening may be provided by adding automatic sprinklers or fireproof shutters to the opening, or by installing the escalator in an enclosed fire-protected hall.
- To limit the danger of overheating, adequate ventilation for the spaces that contain the motors and gears must be provided.
- It is preferred that a traditional staircase be located adjacent to the escalator if the escalator is the primary means of transport between floors

Escalator Working principle

An escalator is a continuously moving staircase. Each stair has a pair of wheels on each side, one at the front of the step and one at the rear. The wheels run on two rails. At the top and bottom of the escalator, the inner rail dips beneath the outer rail, so that the bottom of the stair flattens, making it easier for riders to get on and off



Types of Escalators

a) Moving Walkways

A moving walkway or sidewalk is a conveyor transport system that automatically carries pedestrians. Moving walkways are often installed in pairs for each direction to allow inter-terminal pedestrian travel. They may be used in airports, stadiums, and other large venues where crowds of people want to move in a specific direction.

Moving walkways work by the same basic principle as a conveyor belt: they move at an angle to the ground with little friction. This causes the user less friction from the floor than simply walking on it. The result is a smooth ride without spending energy pushing off the ground.

b) Parallel Escalator

A parallel escalator is an escalator that is side by side with another. The two escalators move in the same direction and at the same speed. This escalator increases capacity, especially in crowded areas like shopping malls or train stations.

The parallel escalator has a few advantages over traditional escalators:

- It can transport more people per unit of time than a straight up-and-down escalator because it has twice as many steps, which are wider than those on a straight up-and-down escalator.
- It can carry more weight because each step can be wider and deeper than those on an up-and-down escalator.
- It uses less space than two straight up-and-down escalators because it doesn't require any room for moving sideways before turning back down again.

c) Stairway Escalators

Stairway escalators are the oldest type of escalator, and they're still used in many places. They're built into a staircase so you can go up or down like a regular staircase. They're also called inclined stairway escalators or inclined moving stairs.

They're especially good for areas where space is limited, like airports and train stations, because they don't take up much space.

d) Belt-Type Escalators

These are the most common types of escalators. The steps are arranged in a rectangular shape, with the outer edges of each step slightly curved. This design allows passengers to stand safely on both sides of the belt. The steps have holes through which the cables run from one end to another and are connected to motors at each end of the escalator.

A typical belt-type escalator has two sets of parallel tracks that run along its length, with a pair of wheels on each track moving up and down as it travels along them. These wheels

move in opposite directions for safety reasons: One set moves forward when going up and backward when going down, while the other set does the opposite.

e) Curved Escalators

The curved escalator is a unique and eye-catching feature in many shopping centers and office buildings. A curved escalator is a moving walkway that is curved in the middle. These escalators are usually found in airports, malls, department stores and other public places.

The **advantage** of curved escalators is that:

They allow for much more efficient use of space than straight escalators. For example, a curved escalator can achieve the same vertical travel with less floor space than a straight one. This is because the radius of the curve allows for an arc longer than the arc created by a straight escalator.

Curved escalators also have some **disadvantages**:

- They are expensive to install.
- They require more maintenance than straight escalators.
- They are not as safe as straight escalators.

f) Wheelchair-Accessible Escalator

Wheelchair-accessible escalators are designed to accommodate wheelchairs and other mobility devices, including walkers, crutches, and canes. They are in high-traffic areas of a facility such as an airport or train station.

Wheelchair-accessible escalators have two sets of steps that alternate in direction as they move up or down. This allows passengers using wheelchairs or other mobility devices to move independently up or down the escalator.

g) Step Type Escalator

Step-type escalators are stair-like devices that transport passengers up or down a vertical distance. They are also referred to as stepped conveyors or simply “steps”.

The most common application of stepped conveyors is the descending type, but some designs also exist for ascending motion. In descending applications, the direction of travel is usually controlled by permanent rules (e.g., you always go down on the left), but some designs allow unidirectional travel with no restrictions on direction (e.g., you can go either way at any given time).

Advantages of Escalators

- Efficient Transportation:** Escalators can move many people between different levels of a building quickly and efficiently, reducing congestion and waiting times, especially in areas with high foot traffic.

- b) **Accessibility:** Escalators provide a convenient and accessible means of vertical transportation for people with mobility challenges, including those using wheelchairs, strollers, or carrying heavy luggage. They often complement elevators in providing accessibility within buildings.
- c) **Space-saving Design:** Compared to traditional staircases, escalators require less horizontal space, making them ideal for areas where space is limited or where a large number of people need to move vertically within a confined space.
- d) **Continuous Movement:** Escalators operate continuously during operating hours, allowing a steady flow of people to move between levels without the need to wait for an elevator or for a staircase to clear.
- e) **Safety Features:** Modern escalators are equipped with various safety features, such as sensors to detect obstructions or irregularities in operation, emergency stop buttons, and handrails to enhance passenger safety and comfort.
- f) **Energy Efficiency:** Escalators are designed to be energy-efficient, with features such as variable speed drives and energy-saving modes that help minimize electricity consumption when not in use or during periods of low traffic.
- g) **Architectural Design:** Escalators can contribute to the aesthetic appeal of a building, especially when integrated into the overall architectural design. They can serve as focal points or design elements that enhance the visual appeal of indoor spaces

Disadvantages of Escalators

- a) **Maintenance Costs:** Escalators require regular maintenance to ensure safe and efficient operation. Maintenance can be costly, involving inspection, lubrication, cleaning, and repairs to mechanical components such as motors, chains, and steps.
- b) **Initial Installation Cost:** The installation of escalators involves significant upfront costs, including equipment purchase, construction, and integration into the building's structure. This can be a barrier to installation, especially for smaller businesses or older buildings without existing escalator infrastructure.
- c) **Energy Consumption:** Escalators consume electricity to operate, and while modern escalators are designed to be energy-efficient, they still contribute to overall energy consumption within a building. Energy usage can be significant, especially in high-traffic areas where escalators run continuously.
- d) **Safety Risks:** Although escalators are generally safe when properly maintained and used correctly, they can pose safety risks, especially for young children, the elderly,

or people with mobility issues. Accidents can occur due to slips, trips, falls, entrapment, or clothing getting caught in moving parts.

- e) **Limited Capacity:** Escalators have a finite capacity determined by their width, length, and speed. During peak periods or in areas with exceptionally high foot traffic, escalators may become overcrowded, leading to congestion and delays.
- f) **Dependency on Electricity:** Escalators rely on electricity to operate, making them vulnerable to power outages or electrical malfunctions. In the event of a power failure, escalators may become temporarily inoperable, requiring alternative means of vertical transportation.

3. TRAVELATORS

Travelators - also known as autowalks, passenger conveyors and moving pavements. They provide horizontal conveyance for people, prams, luggage trolleys, wheelchairs and small vehicles for distances up to about 300 metres. Slight inclines of up to 12° are also possible, with some as great as 18° , but these steeper pitches are not recommended for use with wheeled transport



Types of Travelators

- a) **Horizontal Travelators:** These are the most common type and are typically found in airports, train stations, and shopping malls. They transport passengers horizontally

over flat surfaces and are often used to cover long distances within large buildings or terminals.

- b) **Inclined Travelators:** Inclined travelators are designed to transport passengers up or down slopes or inclines. They are commonly used in areas where there are changes in elevation, such as connecting different levels of a building or moving between platforms in a transportation hub.
- c) **Dual-Directional Travelators:** Dual-directional travelators have two conveyors moving in opposite directions, allowing passengers to travel in both directions simultaneously. They are useful in areas with high pedestrian traffic and can help reduce congestion by accommodating passengers traveling in different directions.
- d) **Multi-Speed Travelators:** Multi-speed travelators offer adjustable speeds to accommodate different levels of pedestrian traffic or user preferences. They may have variable speed settings that can be controlled manually or automatically based on demand.
- e) **Spiral or Circular Travelators:** Spiral or circular travelators are designed in a loop or spiral configuration, allowing passengers to travel in a continuous circular path. These are less common than horizontal or inclined travelators and are typically used in specialized applications or unique architectural designs.
- f) **Outdoor Travelators:** Outdoor travelators are specially designed to withstand exposure to weather elements such as rain, snow, and extreme temperatures. They are commonly used in outdoor environments such as airport terminals, pedestrian walkways, and amusement parks.
- g) **Speed walks:** Speed walks are a type of travelator that operates at a constant speed, typically faster than a walking pace but slower than a moving sidewalk. They are often used in transit systems, exhibition centers, and public spaces where rapid movement is desired.

Features of Travelators

Travelators typically include several features to ensure efficiency, safety, and user convenience. Some common features include:

- a) **Moving Surface:** The travelator consists of a continuous moving surface, usually made of metal slats or rubber belts, which transports passengers along its length. Lengths can range from short stretches to longer distances depending on the location and purpose.

- b) **Handrails:** Handrails are provided on both sides of the travelator to offer stability and support for passengers while they are moving.
- c) **Speed Control:** Travelators often have adjustable speed settings to accommodate different levels of pedestrian traffic and user preferences. Speed control mechanisms ensure smooth acceleration and deceleration.
- d) **Safety Sensors:** Travelators are equipped with sensors to detect obstructions, such as objects or people blocking the path, triggering automatic shutdown or alarms to prevent accidents.
- e) **Emergency Stop Buttons:** Emergency stop buttons are located at regular intervals along the travelator's length, allowing passengers to halt the movement in case of emergencies or safety concerns.
- f) **Directional Signage:** Clear directional signage is provided to indicate the direction of travel and guide passengers to their destination, especially in multi-directional or intersecting travelators.
- g) **Floor Lighting:** Illuminated floor markings or LED lighting along the edges of the travelator provide visibility and enhance safety, particularly in low-light conditions or dimly lit environments.
- h) **Anti-Slip Surface:** The surface of the travelator is designed to be slip-resistant, reducing the risk of accidents or falls, especially in wet or slippery conditions.
- i) **Automatic Start/Stop:** Travelators are equipped with automatic start and stop mechanisms that activate when passengers step onto or off the moving surface, ensuring seamless operation.
- j) **Maintenance Indicators:** Diagnostic systems or indicators may be installed to monitor the condition of critical components and alert maintenance personnel to potential issues or required servicing.
- k) **Accessibility Features:** Travelators are designed to be accessible to individuals with disabilities, with features such as wider pathways, tactile markings, and audible announcements.
- l) **Weather Protection:** Outdoor travelators may include weatherproof enclosures or canopies to protect passengers from rain, snow, and other weather elements.
- m) **Energy Efficiency:** Modern travelators incorporate energy-saving features such as regenerative braking, automatic shut-off, and efficient motor systems to minimize power consumption.

Travelator Operation

- **Activation:** When a passenger steps onto the travelator, sensors detect their presence and initiate the activation sequence.
- **Drive system:** An electric motor powers a series of rollers or wheels that drive the belt at a predetermined speed.
- **Belt design:** The belt is made of multiple layers, including a strong base ply, a traction surface, and a wear-resistant top layer.
- **Control system:** A microprocessor controls the speed, direction, and safety features of the travelator.
- **Maintenance:** Regular maintenance is crucial for safety and smooth operation, including cleaning, lubrication, and inspection of components.

Applications of Travelators

- **Airports:** Travelators are extensively used in airports to help passengers cover long distances between terminals, gates, check-in counters, and baggage claim areas.
- **Train stations:** Train stations often have travelators to facilitate the movement of passengers between different platforms, concourses, and exits.
- **Shopping malls:** Many shopping malls feature travelators to assist shoppers in navigating between different levels or sections of the mall, making it easier to access stores and amenities.
- **Exhibition Centers and Convention Halls:** Large exhibition centers and convention halls often utilize travelators to help attendees move between exhibits, conference rooms, and event spaces.
- **Office Buildings:** Some office buildings incorporate travelators in their lobby areas or interconnected floors to facilitate movement between different office spaces and amenities.
- **Sports Stadiums and Arenas:** Stadiums and arenas may have travelators to assist spectators in reaching their seating areas or accessing various facilities within the venue.
- **Theme Parks:** Theme parks often use travelators to transport visitors between different sections of the park, attractions, and entertainment areas.
- **Hospitals:** In hospitals and medical centers, travelators can aid patients, visitors, and staff in moving between different departments, wards, and treatment areas.

Advantages of Travelators

- **Efficiency:** Travelators help people move quickly and effortlessly over long distances, reducing the time it takes to traverse large public spaces such as airports, train stations, and shopping malls.
- **Accessibility:** Travelators are accessible to people with mobility challenges, including those using wheelchairs, walkers, or strollers. They provide a barrier-free means of transportation within public spaces.
- **Reduced Congestion:** By providing an alternative means of transportation, travelators can help alleviate congestion in crowded areas such as airports and train stations, improving overall flow and efficiency.
- **Convenience:** Travelators allow passengers to transport luggage, shopping bags, and other belongings with ease, eliminating the need to carry heavy loads over long distances.
- **Safety:** Travelators are equipped with safety features such as handrails, sensors to detect obstructions, and emergency stop buttons, enhancing passenger safety and reducing the risk of accidents.
- **Energy Efficiency:** Modern travelators are designed to be energy-efficient, incorporating features such as regenerative braking and automatic shut-off to minimize power consumption when not in use.
- **Cost-Effectiveness:** While initial installation costs may be relatively high, travelators can result in long-term cost savings by reducing the need for additional infrastructure, such as escalators or elevators, and improving operational efficiency.
- **Improved User Experience:** Travelators provide a comfortable and convenient transportation experience for passengers, enhancing their overall satisfaction when navigating public space.

Disadvantages of Travelators

- **Installation and Maintenance Costs:** The initial installation cost of travelators can be high, and they require regular maintenance to ensure smooth operation. This includes servicing motors, replacing belts, and addressing wear and tear on moving parts.
- **Limited Capacity:** Travelators have a limited capacity compared to escalators or elevators, as they typically only accommodate passengers in a single file. During peak periods, this can lead to congestion and longer wait times for users.

- **Space Requirements:** Travelators require a significant amount of space, both horizontally and vertically, which may not always be available in crowded or constrained environments. This can limit their feasibility in certain locations or necessitate extensive modifications to existing infrastructure.
- **Energy Consumption:** While modern travelators are designed to be energy-efficient, they still consume electricity to operate. This ongoing energy consumption can contribute to operating costs and environmental impact.
- **Reliability Issues:** Travelators are complex mechanical systems that are prone to breakdowns and malfunctions, especially with heavy use. This can result in service disruptions, inconvenience for passengers, and increased maintenance costs for operators.
- **Safety Concerns:** While travelators are generally safe when used properly, there is still a risk of accidents or injuries, particularly for children, elderly individuals, and people with mobility impairments. Common hazards include tripping, slipping, or getting clothing or belongings caught in the moving parts.
- **Accessibility Challenges:** Although travelators are designed to be accessible, they may still present challenges for certain individuals, such as those with visual impairments or cognitive disabilities. Ensuring inclusive design and adequate signage is essential to address these concerns.
- **Limited Functionality:** Travelators are primarily designed for horizontal or inclined movement over relatively short distances. They may not be suitable for transporting passengers vertically between different levels, which may require additional infrastructure such as escalators or elevators.

Despite these disadvantages, travelators remain a valuable component of public transportation systems and large public facilities, providing efficient and accessible transportation for millions of people worldwide

4. HOISTS

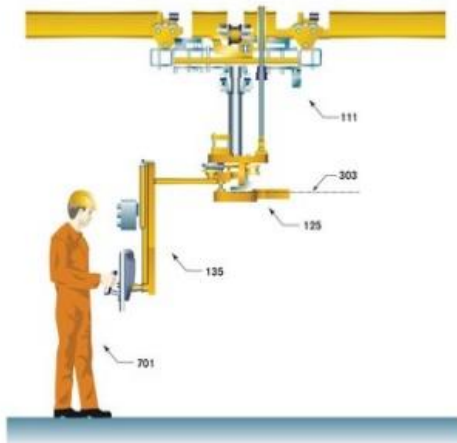
Hoists are mechanical devices used for lifting and lowering heavy loads vertically. In building services, hoists are commonly used for transporting goods, equipment, and sometimes people between different floors of a building. There are various types of hoists, including electric hoists, hydraulic hoists, and manual hoists. They typically consist of a motor or other power source, a drum or pulley system, and a lifting mechanism such as a chain or wire rope. Hoists are essential for efficiently moving materials and equipment within

a building, especially in multi-story structures where manual handling would be impractical or unsafe.

Parts of Hoists

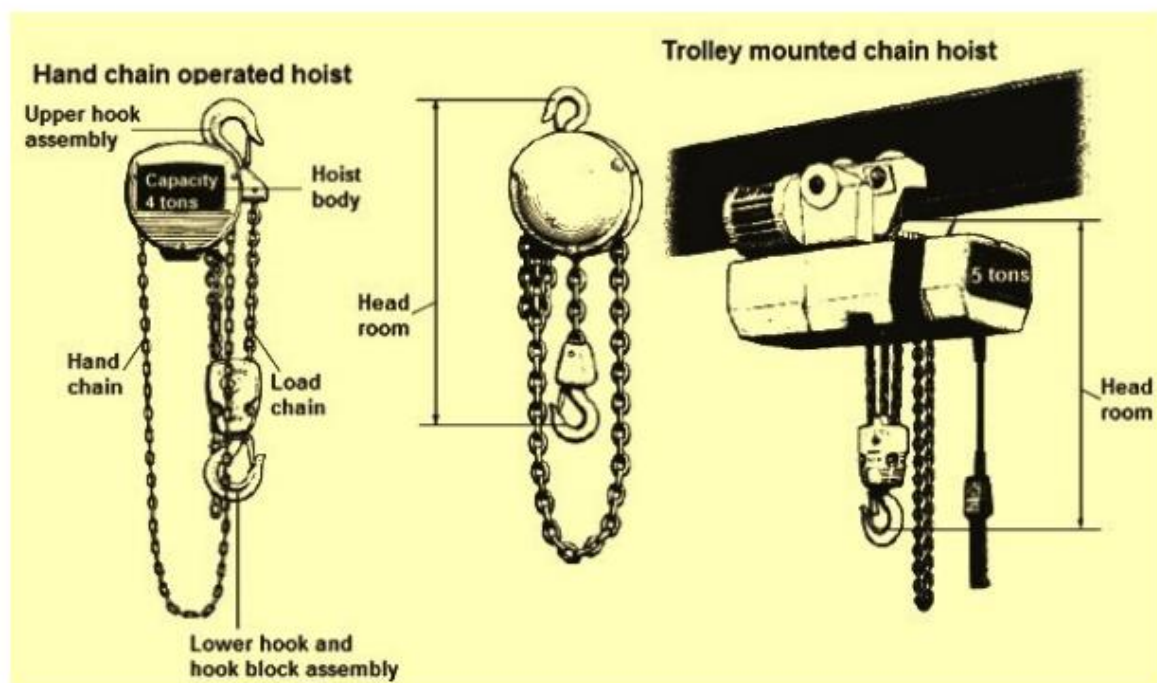
- **Power Source**

This can be an electric motor, hydraulic system, or manual operation.



- **Drum or Pulley System**

This component wraps the lifting mechanism (chain or wire rope) and provides the necessary torque for lifting and lowering loads.



- **Lifting Mechanism**

Consists of a chain or wire rope that connects the load to the hoist. It's responsible for bearing the weight of the load.



- **Control System**

Enables operators to control the movement of the hoist, including lifting, lowering, and stopping.



- **Safety Features**

These include limit switches to prevent over-travel, emergency stop buttons, and overload protection to safeguard against excessive loads.



Functions of Various Parts:

- **Power Source:** Provides the energy needed to operate the hoist.
- **Drum or Pulley System:** Transforms the rotational motion of the power source into linear motion, lifting or lowering the load.
- **Lifting Mechanism:** Bears the weight of the load and transfers it to the hoist for Movement
- **Control System:** Allows operators to manipulate the hoist's movements with precision and safety.
- **Safety Features:** Ensure safe operation by preventing accidents such as overloading or over-travel.

Merits and Demerits of Hoists:

Merits:

1. **Efficiency:** Hoists can efficiently move heavy loads vertically, saving time and labor compared to manual methods.
2. **Safety:** Properly maintained hoists with appropriate safety features can reduce the risk of accidents and injuries.
3. **Versatility:** Hoists come in various types and configurations to suit different applications and load capacities.
4. **Accessibility:** Hoists can transport goods and equipment to different floors, making them accessible in multi-story buildings.
5. **Productivity:** By streamlining material handling processes, hoists can improve overall productivity in building operations.

Demerits:

1. **Initial Cost:** Hoists can be expensive to purchase and install, especially for larger capacity systems or custom installations.
2. **Maintenance:** Regular maintenance is necessary to ensure hoists operate safely and efficiently, which can incur additional costs.
3. **Space Requirements:** Hoists require dedicated space for installation, including overhead clearance for vertical movement.
4. **Limitations:** Hoists may have limitations in terms of load capacity, speed, and travel distance, which can impact their suitability for certain applications.
5. **Complexity:** Depending on the type and configuration, hoists may require specialized training for operation and maintenance, adding complexity to building management.

Understanding these aspects can help in selecting the most suitable hoist for specific building service requirements while considering both the merits and demerits

The Step-by-Step Process of Installation of Lifts and Escalators

Except for the hoist that are installed and used during the construction phase and later removed from site, the rest of the mechanical transportation are useful and some even commissioned post construction.

It is advisable that unless under refurbishment, the choice and designs of lifts, escalator and traveller be done pre-construction to avoid the following challenges;

- Concrete breaking
- Structural adjustments
- Pipe rerouting
- Delays and other cost escalations

The following are the common steps in installation;

Step 1: Site Survey and Traffic Study

Before anything is installed, professionals must:

- Visit your site
- Measure shaft space
- Study foot traffic
- Check power supply
- Confirm number of floors

- Check building use

This step determines: Lift size, Escalator type, Motor power, Safety systems. If this step is done poorly, everything after it suffers.

Step 2: Shaft Construction and Structural Preparation

For lifts, your building must have:

- A proper lift shaft
- Strong concrete walls
- Correct pit depth
- Correct overhead space

For escalators, you must prepare:

- Escalator opening
- Load-bearing beams
- Correct angle support

This stage happens during construction, not after.

Step 3: Electrical and Power Planning

This is very important. Your system may need:

- 3-phase power
- Backup generator
- Automatic changeover
- Surge protection
- Fire system integration

If your power is weak, your system will fail often.

Step 4: Delivery of Lift and Escalator Equipment

Once everything is ready:

- Motors arrive
- Rails arrive
- Cabins arrive
- Escalator trusses arrive
- Control panels arrive

All items are checked before installation starts.

Step 5: Mechanical Installation

For lifts:

- Rails are fixed
- Motor is installed
- Cabin is fitted
- Cables are connected
- Doors are installed

For escalators:

- Truss is installed
- Steps are mounted
- Handrails are fitted
- Drive motor is fixed

This stage requires trained engineers.

Step 6: Electrical Installation and Programming

Here:

- Control boards are programmed
- Floor settings are configured
- Call buttons are tested Emergency systems are activated

This is where your lift becomes “smart.”

Step 7: Testing and Load Trials

Before you use the system, it is tested under:

- Full passenger load
- Emergency power
- Fire mode
- Brake failure simulation

If it fails this stage, it must be corrected immediately.

Step 8: Certification and Handover

Your system is:

- Inspected by authorities
- Certified for use
- Handed over to end user/client
- Maintenance contract begins

For safety purposes, the following are general signs that require immediate attention and you should therefore stop usage:

- Jerky movement

- Strange noise
- Burning smell
- Lift stopping between floors
- Escalator handrail slipping
- System error alarms